

Capstone Project Monthly Progress Report

Month: February 2026

Project Title: Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection

Project ID: CAPSTONE_2022_24250_3

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Total Number of Students in Team: 04

Details of Team Members:

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1 Project Overview

This research work, entitled “**Advancing Biometric Authentication: A Framework for Template Reconstruction Attack Detection**,” offers a safe method to detect fingerprint template reconstruction attacks prior to the attack’s success at compromising biometric authentication systems. The method combines a ResNet-50 deep learning model for robust feature extraction with a hybrid dynamic projection technique to produce secure biometric templates.

Based on the analysis of projection consistency, correlation, and statistical differences between original and reconstructed templates, the system is able to successfully detect malicious attacks while maintaining the efficiency of biometric authentication. The system is tested using the CASIA-FingerprintV5 dataset

2 Work Completed During Last 1 Month

- The development of a secure projection-based fingerprint template protection framework enables the detection of template reconstruction attacks.
- The ResNet-50 model was optimized to produce 512-dimensional fingerprint embedding which achieves strong extraction performance.
- The researchers created a detection system based on projection consistency which employs statistical methods and correlation analysis and distribution analysis methods

- The researchers developed methods to generate simulated reconstruction attacks which included both algorithmic methods and learning-based inversion techniques.
- The researchers conducted comprehensive testing on the CASIA-FingerprintV5 dataset which involved different training and testing scenarios
- The evaluation of system performance used six metrics which included Accuracy and Precision and Recall and F1-score and AUC-ROC and Equal Error Rate (EER).
- Research team documented their experimental results by creating performance visualization graphs which included a confusion matrix and ROC curves and training curves and they used this information to update their research report.

3 Work Planned for this Month

- Extend the analysis of the proposed frameworks to cross-dataset and cross-sensor settings to determine the level of generalization and robustness.
- Develop and analyze the implementation of sophisticated attack plans for reconstruction attacks to further confirm the efficacy of the detection method.
- Optimize the projection-based on the framework to minimize computational complexity in a real-time processing capabilities.
- Compare the proposed framework with existing template protection and attack detection methods as described in the literature.
- Refine the detection thresholds to meet the desired operational False Positive Rates (FPR) with high Genuine Acceptance Rates (GAR).
- Enhance the analysis of the experimental results by including further statistical validation measures.
- Finalize the experimental documentation and work on the manuscript for journal/conference publication.

4 Progress Status

- Overall completion: 80%
- Timeline updated (see Gantt chart if applicable)

5 Challenges/Issues Faced

- Challenge 1 and how it was handled
- Challenge 2 and resolution

6 Learning Outcomes

- Learned tool/technology/algorithm
- Acquired knowledge about XYZ

7 Plan for Next Month

- Task 1
- Task 2
- Expected deliverables

8 Student Individual Contribution

Short reflection (1–2 sentences).

9. Supervisor's Observation

Student Completed: _____% of the planned/assigned task during the last month

Comments:

Number of times students meet with supervisor (Online or Offline): _____

Satisfied with the Student's Work: Satisfied Not Satisfied

Supervisor Signature: _____

Date: DD/MM/YY